SECURITY

DDoS Mitigation Visual Guide

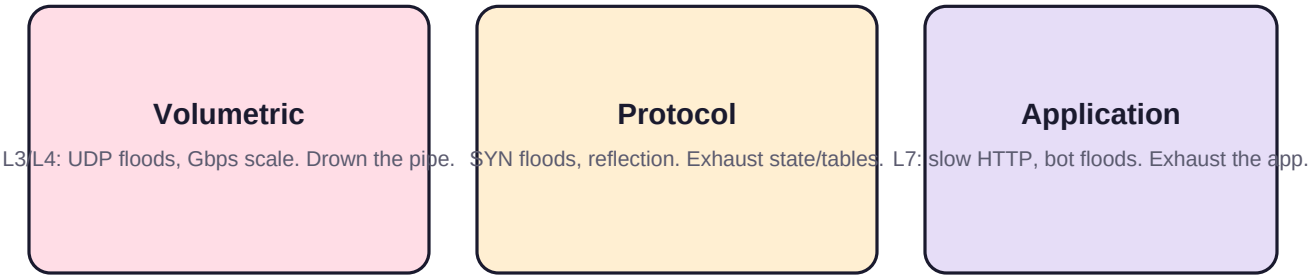
When the internet itself attacks —
layers of defense that absorb the flood.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Three Attack Layers

Different floods need different defenses.

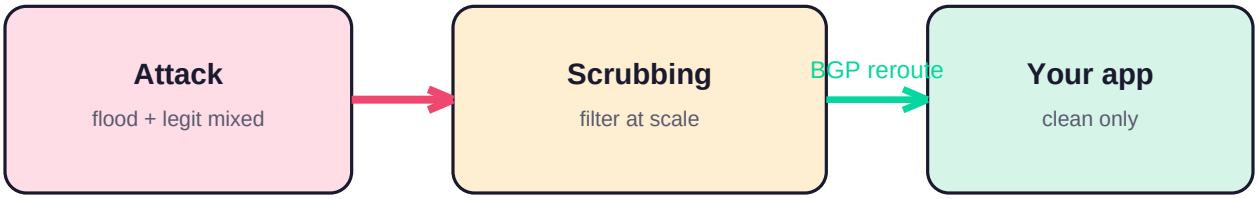


Modern attacks blend all three. Your defense must be layered too — no single box stops everything.

STEP 2/6

Scrubbing Centers

Dirty traffic in, clean traffic out.

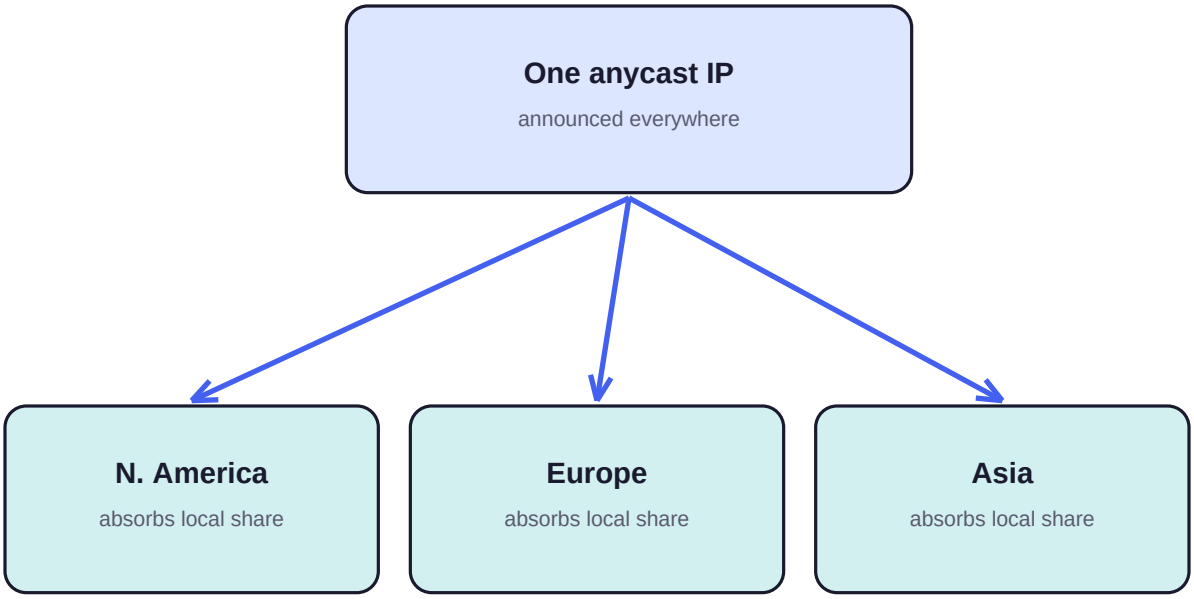


During an attack, BGP routes your prefixes to the scrubber; it drops the junk and tunnels clean traffic back. On-demand or always-on.

STEP 3/6

Anycast Absorption

Spread the flood across the planet.



Attackers in each region hit their nearest PoP — no single link takes the full blast. This is why edge networks matter.

STEP 4/6

Rate-Based & L7 Defense

Application-layer attacks need application-layer answers.

- Rate limit per IP, per session, per endpoint — login and search are favorite targets.
- Challenge suspicious clients: JS challenge, CAPTCHA, proof-of-work before expensive work.
- Cache aggressively — a cached response costs you nothing under flood.
- Bot management: distinguish good bots (search) from bad (scrapers, credential stuffers).

STEP 5/6

Always-On vs On-Demand

Pick your posture before the attack.

Always-on

Traffic always scrubbed. Zero switchover delay; constant cost.

On-demand

Scrubbing activates on detection. Cheaper idle; minutes of pain during spin-up.

High-value targets choose always-on. Everyone else: on-demand WITH tested runbooks and pre-signed BGP authorization.

STEP 6/6

DDoS Readiness Checklist

Do this on a quiet Tuesday, not during the flood.

- Know your baseline: normal pps, bandwidth, request mix per endpoint.
- Pre-authorize traffic rerouting with your provider; test the runbook quarterly.
- Origin protection: never expose origin IPs (DNS history leaks them — rotate after).
- Have a comms plan: status page, customer notice, who declares the incident.

STEP ✓/✓

Key Takeaways

DDoS Mitigation — Remember

- Attacks hit three layers: volumetric, protocol, application — defend all three.
- Scrubbing centers filter floods at scale via BGP rerouting.
- Anycast spreads the blast across regions so no single link drowns.
- L7 needs rate limits, challenges, caching, and bot management.
- Choose always-on vs on-demand deliberately; rehearse the runbook.
- Hide origin IPs and know your traffic baseline before you need it.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev